

Indeterminate Forms

Find the limit using L' Hospital's rule:

$$\begin{aligned} & 1. \lim_{x \rightarrow 1} \frac{\ln x}{x-1}, \quad 2. \lim_{x \rightarrow 3} \frac{x-3}{3x^2-13x+12}, \quad 3. \lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \frac{\cos 3x}{x^2} \right) \quad 4. \lim_{x \rightarrow \pi} \frac{\sin x}{x-\pi} \\ & 5. \lim_{x \rightarrow 0} \frac{x - \tan^{-1} x}{x^3}, \quad 6. \lim_{x \rightarrow +\infty} \frac{e^{3x}}{x^2}, \quad 7. \lim_{x \rightarrow 0} \frac{a^x - 1 - x \log a}{x^2}, \quad 8. \lim_{x \rightarrow 0} (e^x + x)^{\frac{1}{x}} \\ & 9. \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right) \quad 10. \lim_{x \rightarrow \pi} (x - \pi) \cot x, \quad 11. \lim_{x \rightarrow 0} \frac{\ln(\sin x)}{\ln(\tan x)}, \quad 12. \lim_{x \rightarrow \infty} x e^{-x} \\ & 13. \lim_{x \rightarrow 0} \frac{\sin 2x}{x}, \quad 14. \lim_{x \rightarrow 0} \frac{\sin x}{x^2}, \quad 15. \lim_{x \rightarrow \infty} \frac{x}{e^x}, \quad 16. \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{x e^x} \right) \quad 17. \lim_{x \rightarrow 0} \frac{\sin 2x}{\sin 5x}. \end{aligned}$$

Answers:

1. 1

2. $\frac{1}{5}$

3. $\frac{9}{2}$

4. -1

5. $\frac{1}{3}$

6. $+\infty$

7. $\frac{(\log a)^2}{2}$

8. e^2

9. 0

10. 1

11. 1

12. 0

13. 2

14. $+\infty$ for $x > 0$

$-\infty$ for $x < 0$

Two sided limit does not exist.

15. 0

16. 1

17. $\frac{2}{5}$